Individual Reflection

Mini-Project 01: Customers Like You | Individual submission

What I did

Working independently, I took responsibility for both the data engineering and machine learning parts of this project. I cleaned the two years of retail transactions, removed invalid or non-purchase records, and used a time-based split to keep future purchases out of training. I built a sparse customer–product matrix, applied Truncated SVD and PCA, evaluated top-10 product recommendations against a best-seller baseline, and visualized four customer segments. I also organized the scripts and notebook so the analysis could be reproduced.

What I learned

The main lesson was that a smaller and faster model is not automatically a more useful one. With 50 components, SVD produced a compact representation of purchase behavior, but its hit rate at 10 was **24.05%**, below the best-seller baseline of **26.87%**. PCA performed similarly at **24.10%**. I initially expected the latent factors to find better product relationships, so this result showed me why testing on unseen future purchases and comparing against a simple baseline are essential. I also learned that compression has a cost: reducing the matrix to a few components loses information about individual buying patterns.

What I would change

With more time, I would use a separate validation period to select the number of components and compare purchase-weighting choices, without tuning on the final test data. I would also test a hybrid recommender that combines product similarity with popularity, examine whether results change across seasons, and validate the customer-segment labels using their actual product and country distributions. Finally, I would benchmark the full nightly pipeline, including data loading and saving outputs, rather than only the model computations.